

# Sound Manager Implementation

*Sanguine Echoes — Audio System Summary*

Updated: August 12, 2026 • Phaser 4 + TypeScript Strict • Vite

## ◆ 1. Core Service Created

**File:** `src/services/SoundManager.ts` — instantiated per scene ( `MainMenuScene` , `LevelScene` , `SanctuaryScene` , `TreasureScene` ) and exposed as `scene.soundManager` .

- **Master / BGM / SFX** volume + mute controls, each exposed as a getter/setter
- **localStorage persistence** — volumes and mute states survive across sessions
- **Anti-overlap throttling** — min interval per SFX key (default 50ms; footsteps 80ms)
- **BGM fade-in/out** — `playBGM(key, loop, fadeInMs)` and `stopBGM(fadeOutMs)` ; a new track stops the previous one
- **Missing-key guard** — logs a warning instead of throwing if a key isn't in the audio cache
- **Auto-cleanup** — SFX destroy themselves on `complete` ; `stopAllSFX()` fires on player death
- **Pause support** — `pauseAll()` / `resumeAll()` wired to the in-game pause menu, plus `pauseBGM()` / `resumeBGM()`
- **Random-variant helper** — `playRandomSFX(baseKey, count)` for future footstep/jump variant pools

## ◆ 2. Audio Asset Keys Registered

**31 real assets** in `src/assets/audio/` (~17MB, `.wav` + `.mp3` , imported via Vite).  
Registered per scene: `LevelScene` loads 25 keys, `MainMenuScene` 5, `SanctuaryScene` 4, `TreasureScene` 1.

| CATEGORY         | KEYS (COUNT)                                                                                                                                                                                            |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BGM              | bgm_level1 , bgm_level2 , bgm_level3 , bgm_main_menu ,<br>bgm_sanctuary (5)                                                                                                                             |
| PLAYER           | sfx_jump , sfx_dash , sfx_attack_swish , sfx_plunge ,<br>sfx_charging_attack , sfx_fully_charged , sfx_parry ,<br>sfx_parry_hit , sfx_surge , sfx_footstep , sfx_player_hurt ,<br>sfx_player_death (12) |
| ENEMY &<br>BOSS  | sfx_enemy_hit , sfx_enemy_die , sfx_monster_attack ,<br>sfx_enemy_particle , sfx_enemy_fireball_explosion ,<br>sfx_boss_roar , sfx_boss_melee_attack ,<br>sfx_boss_shoot_particle (8)                   |
| UI               | sfx_ui_hover , sfx_ui_confirm , sfx_ui_declined ,<br>sfx_ui_pause_menu (4)                                                                                                                              |
| WORLD &<br>ITEMS | sfx_echo_pickup , sfx_open_chest (2)                                                                                                                                                                    |

### ◆ 3. Sound Hooks Implemented

| CATEGORY | EVENT                                 | SOUND KEY        | LOCATION     |
|----------|---------------------------------------|------------------|--------------|
| Player   | Jump /<br>air re-<br>jump             | sfx_jump         | jumpState    |
|          | Dash (L)                              | sfx_dash         | dashState    |
|          | Melee<br>swings<br>(combo /<br>heavy) | sfx_attack_swish | playerStates |
|          | Plunge<br>dive                        | sfx_plunge       | plungeState  |

| CATEGORY     | EVENT                | SOUND KEY                       | LOCATION                    |
|--------------|----------------------|---------------------------------|-----------------------------|
|              | Charge start         | sfx_charging_attack             | chargingState               |
|              | Charge fully charged | sfx_fully_charged               | LevelScene                  |
|              | Parry window (K)     | sfx_parry                       | parryState                  |
|              | Parry counter        | sfx_parry_hit                   | parryCounterState           |
|              | Ultimate (Surge)     | sfx_surge                       | ultimateState               |
|              | Footsteps            | sfx_footstep                    | AshenHunter (80ms throttle) |
|              | Player hurt          | sfx_player_hurt                 | AshenHunter.takeHit()       |
|              | Player death         | sfx_player_death + stopAllSFX() | deadState                   |
| Enemy & Boss | Enemy hit            | sfx_enemy_hit                   | Enemy.takeHit()             |
|              | Enemy death          | sfx_enemy_die                   | deadState                   |
|              | Melee enemy attack   | sfx_monster_attack              | attackState                 |
|              | Projectile launch    | sfx_enemy_particle              | EnemyProjectile             |

| CATEGORY      | EVENT                 | SOUND KEY                    | LOCATION                                                |
|---------------|-----------------------|------------------------------|---------------------------------------------------------|
|               | Fireball explosion    | sfx_enemy_fireball_explosion | EnemyProjectile                                         |
|               | Boss roar (intro)     | sfx_boss_roar                | LevelScene                                              |
|               | Boss melee            | sfx_boss_melee_attack        | HellBeast / DragonBoss / ArchDemonBoss                  |
|               | Boss projectile       | sfx_boss_shoot_particle      | DragonBoss / ArchDemonBoss                              |
| World & Items | Echo pickup / reclaim | sfx_echo_pickup              | EchoDrop / DeathMarker                                  |
|               | Open chest            | sfx_open_chest               | TreasureScene                                           |
| UI            | Hover                 | sfx_ui_hover                 | MenuList / ControlsPanel / CreditsPanel                 |
|               | Confirm / accept      | sfx_ui_confirm               | MenuList / SettingsPanel / ControlsPanel / CreditsPanel |
|               | Decline / back        | sfx_ui_declined              | MenuList / ControlsPanel                                |
|               | Pause menu            | sfx_ui_pause_menu            | PauseMenu / ConfirmModal / CreditsPanel                 |

◆ 4. BGM Playback

| SCENE                | TRACK         | FADE-IN | NOTES                             |
|----------------------|---------------|---------|-----------------------------------|
| Main Menu            | bgm_main_menu | 2s      | Fades out (400ms) on scene change |
| L1 Gallows Graveyard | bgm_level11   | 2s      | Per-level track                   |
| L2 Haunted Forest    | bgm_level12   | 2s      | Per-level track                   |
| L3 Night Town        | bgm_level13   | 2s      | Per-level track                   |
| Sanctuary (hub)      | bgm_sanctuary | 2s      | —                                 |
| Victory              | — (none)      | —       | Silent; track pending             |

Every `playBGM` call stops the previous track first, so scene transitions never overlap audio. The pause menu calls `pauseAll()` and unpausing calls `resumeAll()` , keeping BGM and SFX in sync with the game state.

◆ 5. Files Modified / Created

|                                                                                                            |                                                                          |
|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| <b>src/services/SoundManager.ts</b><br>Created                                                             | <b>src/assets/audio/</b><br>Created — 31 Audio Files                     |
| <b>src/scenes/LevelScene.ts</b><br>Modified — Preloads 25 Keys, Per-Level BGM, Boss Roar, Pause/Resume All | <b>src/scenes/MainMenuScene.ts</b><br>Modified — Menu BGM + UI SFX       |
| <b>src/scenes/SanctuaryScene.ts</b><br>Modified — Hub BGM + UI SFX                                         | <b>src/scenes/TreasureScene.ts</b><br>Modified — Chest-Open SFX          |
| <b>src/entities/player/playerStates.ts</b><br>Modified —<br>Jump/Dash/Attack/Plunge/Charge/Parry/Surge/De  | <b>src/entities/player/AshenHunter.ts</b><br>Modified — Footsteps + Hurt |

**src/entities/enemies/Enemy.ts**

Modified — Enemy Hit

**src/entities/enemies/enemyStates.ts**

Modified — Melee Attack + Death

**src/entities/enemies/EnemyProjectile.ts**

Modified — Launch + Explosion

**src/entities/enemies/ (bosses)**Modified — HellBeast, DragonBoss,  
ArchDemonBoss (+ Fire-Skull / Ghost /  
Spellcaster Projectile Spawners)**src/entities/pickups/**Modified — EchoDrop + DeathMarker Pickup  
SFX**src/ui/**Modified — SettingsPanel (Volume/Mute),  
MenuList, PauseMenu, ConfirmModal,  
ControlsPanel, CreditsPanel,  
GemChoicePanel, UpgradePanel

**Note:** All audio is now real — 31 files ( .wav + .mp3 ) live in  
src/assets/audio/ and are imported through Vite, so they're bundled with  
the build. Credit/attribution for the packs is tracked in the project credits (Mini  
Fantasy Dungeon, 400 Sounds Pack, Universal UI SFX, RPG Essentials SFX — all  
free/CC0).

## ◆ 6. Verification

✓ TypeScript typecheck passes

✓ Vite build passes

All 31 audio assets registered. Sounds hooked to every core gameplay event —  
player states, enemies & bosses, pickups, UI, and per-scene BGM.

Sanguine Echoes — 2D Gothic Hack-and-Slash Roguelite  
Phaser 4 + TypeScript Strict + Vite